AQA GCSE Physics
Topic – Waves
Section 1: Detailed Revision Notes

Types of Waves: Transverse vs Longitudinal

What is a Wave?

A wave is a **transfer of energy** from one place to another **without transferring matter**.

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1. Transverse Waves

In a **transverse wave**, the vibrations are **perpendicular (at right angles)** to the direction of energy transfer.

Examples:

- Light waves
- Water waves
- All electromagnetic waves

Key Features:

- Have **crests** (highest point)
- Have **troughs** (lowest point)

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2. Longitudinal Waves

In a **longitudinal wave**, the vibrations are **parallel** to the direction of energy transfer.

Example:

- Sound waves

Key Features:

- **Compressions** (particles close together)
- **Rarefactions** (particles spread apart)

Wave Properties & Key Definitions

Oscillation

One complete vibration or movement.

Rest Position

The central position where no vibration occurs.

Amplitude

Maximum displacement from the rest position.

Wavelength (λ)

Distance from one identical point on a wave to the next identical point (e.g. crest to crest).

Frequency (f)

Number of waves passing a point per second.

Unit: **Hertz (Hz)**

Period (T)

Time taken for one complete wave.

Measuring Waves: Wavelength, Frequency, Amplitude

Measuring Wavelength

- Measure distance between two consecutive crests or compressions.

Measuring Amplitude

- Measure from rest position to crest (or trough).

Measuring Frequency

Count number of waves in a given time.

Frequency formula:

f = number of waves ÷ time

Wave Equation & GCSE Calculations

The Wave Equation

$v = f \times \lambda$

Where:
v = wave speed (m/s)
f = frequency (Hz)
λ = wavelength (m)

Rearranged Forms

$f = v \div \lambda$

$\lambda = v \div f$

Example Calculation

A wave has frequency 5 Hz and wavelength 2 m.

$v = 5 \times 2$
$v = 10$ m/s

Reflection, Refraction and Diffraction

Reflection

When a wave bounces off a surface.

Angle of incidence = angle of reflection.

Refraction

When a wave changes direction because it changes speed entering a new medium.

Frequency stays the same.
Wavelength changes.

Diffraction

Spreading out of waves when they pass through a gap or around an obstacle.

More diffraction occurs when:

- Gap size is similar to wavelength
- Wavelength is large

Sound Waves & Human Hearing

Sound is a **longitudinal wave**.

Produced by vibrating objects.

Speed of Sound

In air: approximately **340 m/s**

Sound travels faster in solids than liquids, and slowest in gases.

Human Hearing Range

20 Hz to 20,000 Hz

Below 20 Hz → Infrasound
Above 20,000 Hz → Ultrasound

Pitch and Volume

Pitch depends on frequency.
Volume depends on amplitude.

Required Practical: Investigating Waves

Students must be able to:

- Use a ripple tank or string to observe waves
- Measure wavelength
- Calculate wave speed
- Rearrange wave equation
- Calculate frequency

Steps:

1. Measure wavelength using ruler.
2. Count waves in given time to find frequency.
3. Use wave equation to calculate speed.

Electromagnetic Spectrum Overview

All EM waves are **transverse waves**.
All travel at the same speed in a vacuum:

3×10^8 m/s

Order of EM Spectrum (Low → High Frequency)

Radio waves
Microwaves
Infrared
Visible light
Ultraviolet
X-rays
Gamma rays

EM Spectrum Uses & Hazards

Radio Waves

Communication

Microwaves

Cooking, satellite communication

Infrared

Heaters, thermal imaging

Visible Light

Seeing

Ultraviolet

Fluorescent lamps, tanning
Hazard: Skin damage

X-rays

Medical imaging
Hazard: Cell damage

Gamma Rays

Cancer treatment
Hazard: Ionising radiation

Ionising radiation can cause mutations and cancer.

GCSE Waves Formula & Unit Summary

Wave speed equation:

$v = f \times \lambda$

Units:
Wave speed → m/s
Frequency → Hz
Wavelength → m

Period equation:

$T = 1 \div f$

Units:
Period → seconds (s)

Common GCSE Exam Mistakes

- Forgetting to convert cm to metres
- Confusing amplitude and wavelength
- Changing frequency during refraction (frequency stays constant)
- Writing wrong units
- Mixing transverse and longitudinal descriptions

End of Section 1 – Waves Revision Notes